THE AUDIO FORMULA: PREDICTING HIT SONGS USING MACHINE LEARNING

COSC 5360 Midterm Project

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INTRODUCTION.

Music streaming platforms generate massive amounts of data.

Understanding what makes a song a “hit” can help:

- **RECORD LABELS** identify potential hits.
- **STREAMING PLATFORMS** improve recommendations.
- **ARTISTS** understand characteristics of successful songs.

LITERATURE REVIEW.

- Previous studies on hit song prediction often rely on combining audio features with external factor like **SOCIAL MEDIA** activity to improve accuracy.
- Existing research utilizing the Spotify Web API shows that raw acoustic properties hold predictive power, but models can have weak explanatory power or limited linear correlation due to human factors of music.
- **THE GAP:** This project builds on prior research by isolating the raw **"AUDIO FORMULA"** and optimizing a Random Forest classification model - a method proven highly successful in similar Spotify prediction studies - specifically for high-stakes industry application.

PROBLEM FORMULATION.

- **The Question:** Is commercial music success purely subjective, or is there a **MEASURABLE PATTERN**?
- **The Goal:** Build a **BINARY CLASSIFICATION** model to act as a **HIT PREDICTOR** based on the audio properties of a song.
- **Target Variable:** `Is_hit`
  - **0** = Normal Song
  - **1** = Hit Song
- **Key Challenge:** The dataset is **HIGHLY IMBALANCED** because only a small percentage of songs become hits.

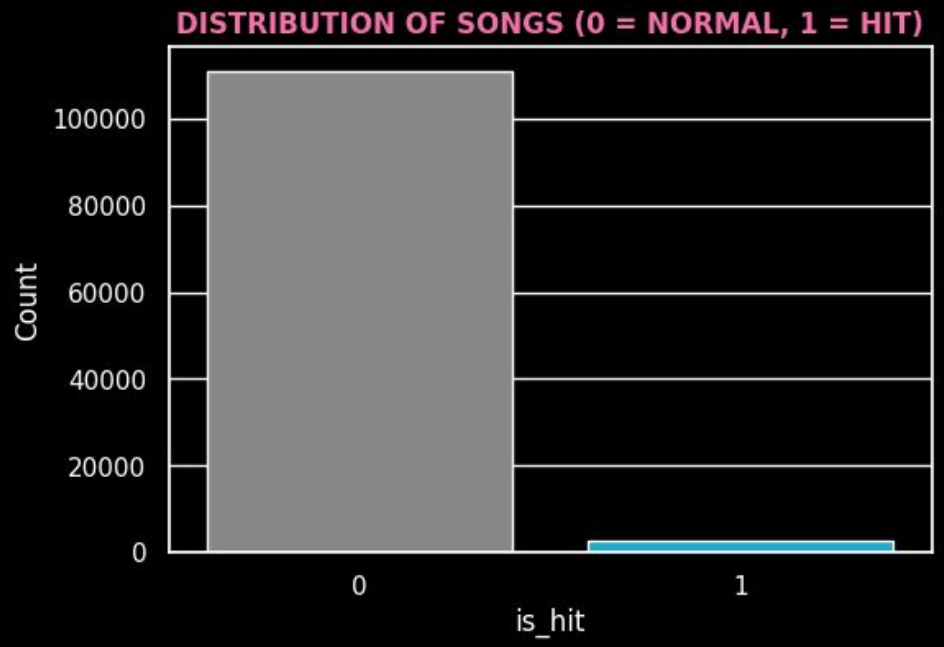
THE SPOTIFY TRACKS DATASET.

- Analyzed over 114,000 tracks sourced from Kaggle.
- Extracted audio features acting as variables:
  - ***DANCEABILITY:*** Suitability for dancing.
  - ***ENERGY:*** Intensity and activity.
  - ***LOUDNESS:*** Average loudness of the track.
  - ***ACOUSTICNESS:*** Whether the song is acoustic.
  - ***VALENCE:*** Musical positiveness (happiness).
  - ***TEMPO:*** Beats per minute (BPM).
- After cleaning 113,999 songs remained.

MODEL IMPLEMENTATION STEPS.

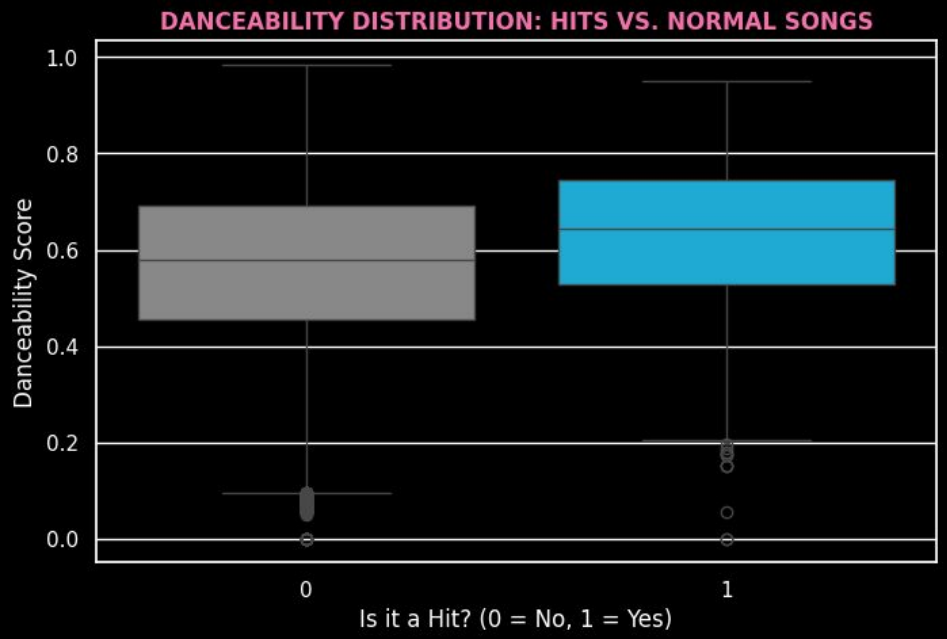
- Data loading and cleaning.
- **FEATURE SELECTION.**
- **TRAIN/TEST SPLIT** (80% training, 20% testing).
- Model training.
- Model evaluation.
- **HYPERPARAMETER TUNING** (GridSearch).
- **FINAL MODEL EVALUATION.**

DATA DISTRIBUTION.



- The dataset is **HEAVILY IMBALANCED** (Hits are rare, which mirrors the real-world music industry).

DOES DANCEABILITY MATTER?



- Initial visual analysis shows a slight shift in distribution, indicating that hits often lean toward **HIGHER DANCEABILITY SCORES**.

LOGISTIC REGRESSION.

- Built as our baseline **LINEAR CLASSIFICATION** model
- Utilized a **STANDARD SCALER** pipeline to perfectly normalize the varying audio features (tempo and loudness) before training.
- Simple linear approach struggled to capture the complexity of what make a song popular.

RANDOM FOREST CLASSIFIER.

- Applied a **RANDOM FOREST** model to handle complex, non-linear interactions between the audio features.
- Used an **80/20 TRAIN-TEST SPLIT** to ensure the model was evaluated on unseen data.
- Applied **GRID SEARCH** to optimize hyperparameters and maximize the model's accuracy.

MODEL EVALUATION.

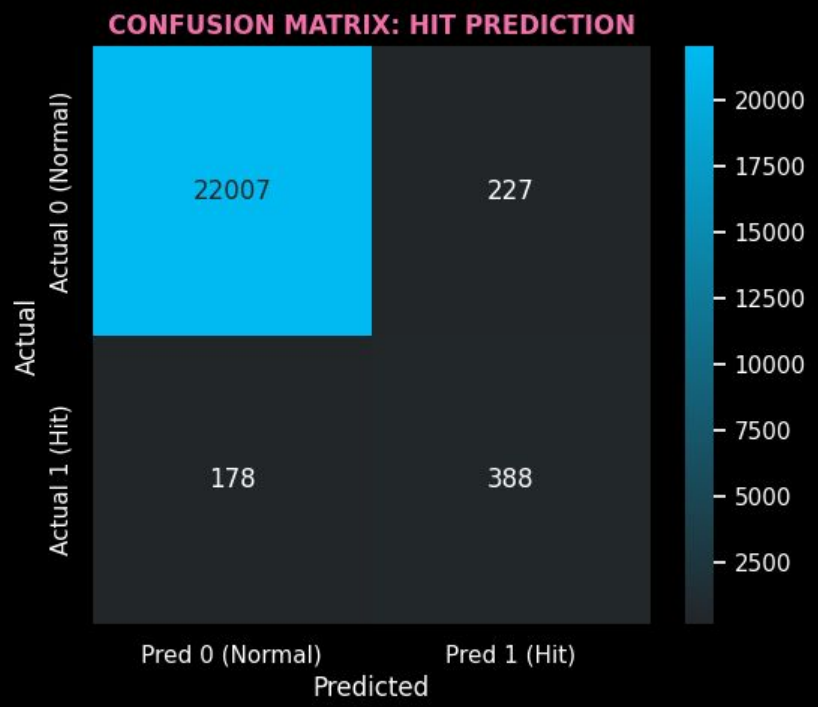
LOGISTIC REGRESSION (BASELINE)

- Accuracy: 59.46%
- ROC-AUC: 0.7263
- Performance was restricted due linear model not handling highly imbalanced dataset.

RANDOM FOREST (OPTIMIZED)

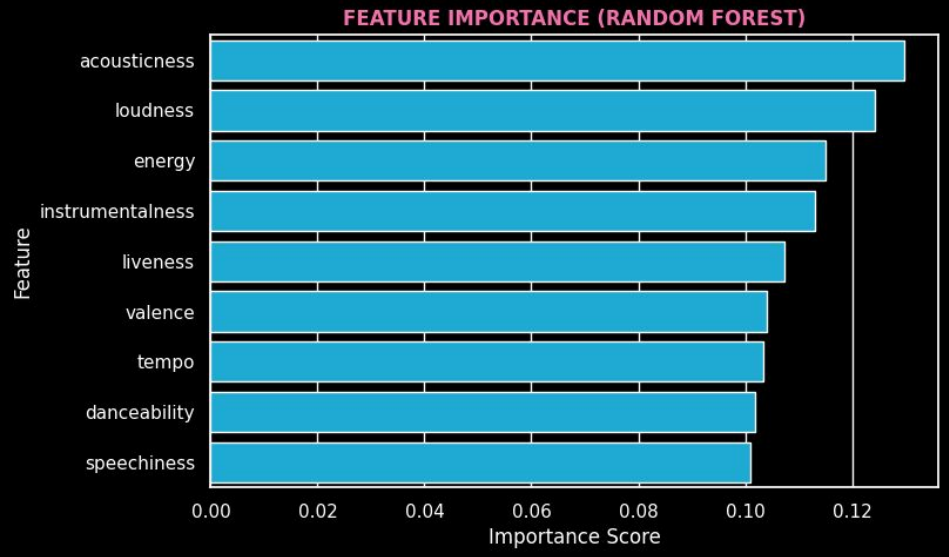
- Accuracy: 98.22%
- ROC-AUC: 0.9061
- Significantly improved predictions performance by mapping non-linear relationships between raw audio features.

MODEL ACCURACY.



- **98.22%**
- The confusion matrix highlights the model's precision in **MINIMIZING FALSE POSITIVES** when predicting hits.
- This ensures that when the model predicts a song is a hit, it is highly accurate, which is exactly what a record label would want.

THE AUDIO FORMULA: FEATURE IMPORTANCE.



- Our Random Forest model identified exactly which audio properties carry the most weight when predicting a commercial hit.

CONCLUSION & REAL WORLD APPLICATION.

- **MUSIC ISN'T JUST ART; IT IS DATA.** Machine learning can successfully identify the underlying audio formulas that resonate with the masses.
- **Application:** Artists and producers (like **HARRY STYLES**) could theoretically run demo tracks through this model to see if the audio chemistry aligns with historical hit data before officially releasing a lead single or album.

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5. Spotify Tracks Dataset. Kaggle.

   https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset